SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Akshitha / 192521390 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Akshitha.S
Signature of the student:

1. Event Communication security

a) Reconstruct the encrypted message using a fixed shift of 4 positions.

The original message is:

MEET AT NOON

using a caesar cipher with shift +4

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

T	U	V	W	X	Y	Z
19	20	21	22	23	24	25

Encryption:

$$C = (P + K) \bmod 26$$

M=12, $C = (12+4) \bmod 26 = 16 = Q$

$(4+4) \bmod 26 = 8 = I$

$(4+4) \bmod 26 = 8 = I$

$(19+4) \bmod 26 = 23 = X$

$(0+4) \bmod 26 = 4 = E$

$(19+4) \bmod 26 = 23 = X$

$(13+4) \bmod 26 = 17 = R$

$(14+4) \bmod 26 = 18 = S$

$(14+4) \bmod 26 = 18 = S$

$(13+4) \bmod 26 = 17 = R$

MEET AT NOON ⟶ QIIX EX RSSR

b) Decryption

$$P = (C - K) \bmod 26$$

Here, K=4

For example,

Q = 16

$$P = (6 - 4) \bmod 26 = 12$$

$12 = M$

Thus

2111X EX RSSR $\longrightarrow$ MEET AT NOON

5. Legacy Encryption in Corporate Communication

Given Ciphertext:

"WECRLTEERDSOEEFEAOCAIVDEN"

The system uses a 3-level transposition arrangement, which can be treated as a 3-row rail Fence transposition

a) Determine the most probable original message.

Reversing the 3-level transposition gives:

WECRLTEERDSOEEFEAOCAIVDEN

WE ARE DISCOVERED FLEE AT ONCE

Therefore, the original message is:

"WE ARE DISCOVERED FLEE AT ONCE"

b) Why is this method vulnerable in modern communication systems?

A transposition cipher only changes the positions of the letters; it does not change the letter themselves.

It is vulnerable because:

1. It has a small number of possible

arrangements

2. An attacker can easily try different arrangements

3. Frequency and letter patterns remain unchanged

4. Modern computers can quickly test possible permutations.

5. It does not provide strong protection against modern cryptanalysis.

c) One practical enhancement

The company should replace the old transposition method with a modern encryption algorithm such as AES-256.

AES provides much stronger confidentially because it uses a large key space and multiple rounds of encryption, making brute-force and pattern-based attacks extremely difficult.